

AUTOMATED FILE ORGANIZER

Algorithm and Dataset

Algorithm

Step-by-Step Procedure

1. Initialize the application and load the required Python modules.
2. Allow the user to select a source directory containing files.
3. Retrieve all files available in the selected directory.
4. Validate each item to ensure it is a file and not a folder.
5. Extract the file extension and compare it with predefined classification rules.
6. Determine the appropriate destination category for each file.
7. Automatically create category folders if they are unavailable.
8. Relocate files to their respective folders without modifying their contents.
9. Continue the process until every eligible file has been categorized.
10. Generate a completion report summarizing the organization process and terminate the application.

Logic / Method Used

The application adopts an extension-driven classification strategy. Each file is examined based on its extension, mapped to a predefined category, and transferred to the corresponding destination folder. The process minimizes manual effort while ensuring consistent and systematic file organization.

Model / Technique Applied

- Rule-Based File Categorization
- Extension Mapping Technique
- Automated Directory Management
- Python File Handling (os and shutil modules)
- Sequential File Processing

Process Flow of the System

Start → Initialize System → Select Source Directory → Scan Available Files → Detect File Extension → Apply Classification Rules → Create Category Folder (if required) → Move File → Update Process Status → Generate Final Report → End

Dataset

Dataset Description

The project uses a real-time collection of heterogeneous files located in a user-selected directory. Instead of relying on a fixed machine learning dataset, the system processes live files of different formats to evaluate the efficiency and accuracy of automated file organization.

Input Dataset

Document files (.pdf, .doc, .docx, .txt, .pptx, .xlsx), image files (.jpg, .jpeg, .png, .gif), video files (.mp4, .avi, .mkv), audio files (.mp3, .wav), compressed files (.zip, .rar, .7z), and miscellaneous file types.

Output Dataset

After execution, the system automatically places files into categorized folders such as Documents, Images, Videos, Audio, Archives, and Others, producing a well-structured directory for efficient file management.